

BUS TOPOLOGY

1. Project Overview

This document describes the design, configs and implementation of a local area network using a logical bus topology that is built with the help of Cisco Packet Tracer. In this project the principle of the bus is greatly demonstrated especially because all the devices are placed on the same VLAN (10), and all the devices communicate using a single communication channel creating with daisy-chained switches. In this topology I made use of both wired and wireless clients.

- Project Title: Bus_PathFinder
- Objective: Demonstrate how a bus works by creating a single broadcasting domain network in which all devices communicate through the same backbone channel both wired and wireless.
- Topology: Bus Topology
- Core Tech: VLAN

2. Materials used:

Item	Quantity	Model
Access Switch	5	2930 24TT Switch
Access Point	1	Access-Point PT-AC
Tablet	1	TabletPC-Pt
Smartphone	1	Smartphone-PT
Server	1	Server-PT
PC	1	PC-PT
Printer	2	Printer-PT
Router	1	Router_PT
Laptops	2	Laptop-PT

3. Network design and topology

Logical topology diagram:

Logically the network is performing exactly the same as a bus is expected to. By placing all the switches in a daisy-chain the connected end devices get to mimic being connected to a backbone cable where all the data being sent is broadcasted to all the devices that are connected.

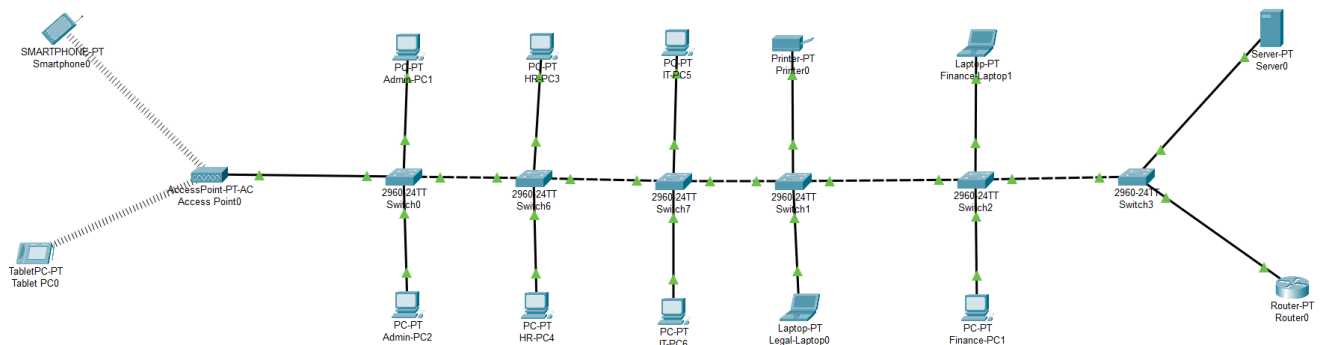


Diagram Explanation:

- The daisy-chained switches form the bus/backbone of the device in which the info will mainly be moving, switch 1 connected to switch 2, switch 3 connected to switch 4 and so on until it reaches the end and one of the switches has an access point connected to switch 1.
- All the end devices are connected to both the switches and the access point directly and info can flow through this connection.
- Wireless integration: One of the switches has the access point connected to it and thus introduces a pathway for the wireless devices to join the network.

4. Configurations:

The aim of this design was simplicity and thus the switches did not to be configured explicitly since they are all operating on the same VLAN, therefore, in this case their only duty was to forward the packets thus acting as a backbone.

- Access Point Configuration
 - Operation mode: Set to “Access Point (0)”
 - SSID: SecureTrustWifi

- Wireless security:WPA2-PSK
- Client device config:
Wireless devices connect to the SSID and enter the pass key
“Network123”

5. IP addressing:

The devices on the network were all assigned IP addresses through the use of the server with DHCP service enabled thus giving them out automatically.

VLAN: 10

Network address: 10.0.0.0/24

Subnet mask: 255.255.255.0

Default gateway: 10.0.179.0

DNS: 10.0.179.0

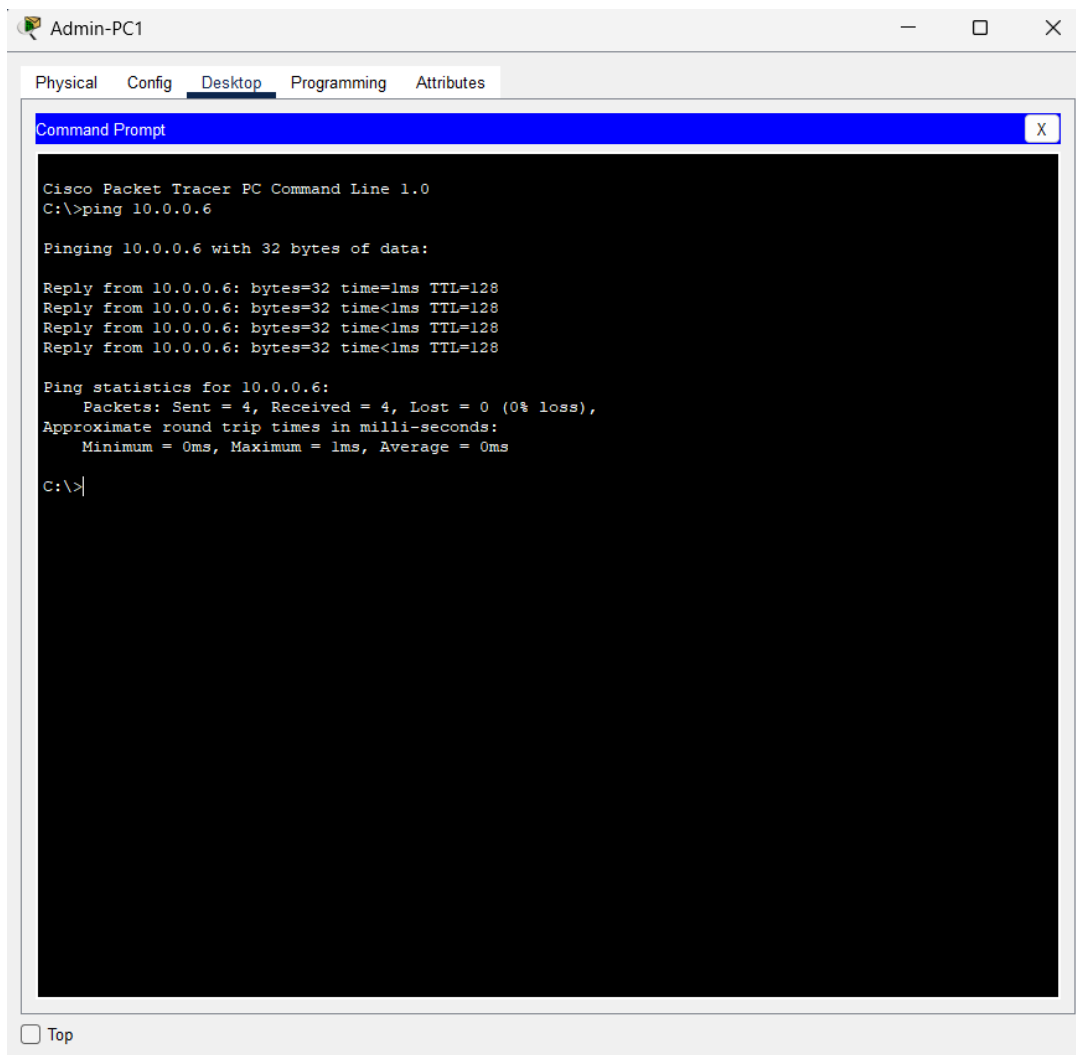
Device	Type	IP address	VLAN	Remark
Admin-Pc1	Wired	10.0.0.12	10	Connected to switch 0
Admin-Pc2	Wired	10.0.0.11	10	Connected to switch 0
HR-Pc3	Wired	10.0.0.15	10	Connected to switch 6
HR-PC4	Wired	10.0.0.10	10	Connected to switch 6
IT-PC5	Wired	10.0.0.8	10	Connected to switch7
IT-PC 6	Wired	10.0.0.9	10	Connected to switch7
Smartphone	Wireless	10.0.0.16	10	Connected to access point

Tablet	Wireless	10.0.0.17	10	Connected to access point
Legal-laptop0	Wired	10.0.0.6	10	Connected to switch 1
Printer0	Wired	10.0.179.0	10	Connectd to switch 1
Finance-PC1	Wired	10.0.0.7	10	Connected to switch 2
Finance-laptop1	Wired	10.0.0.14	10	Connected to switch 2
Server	Wired	10.0.179.1	10	Connected to switch3

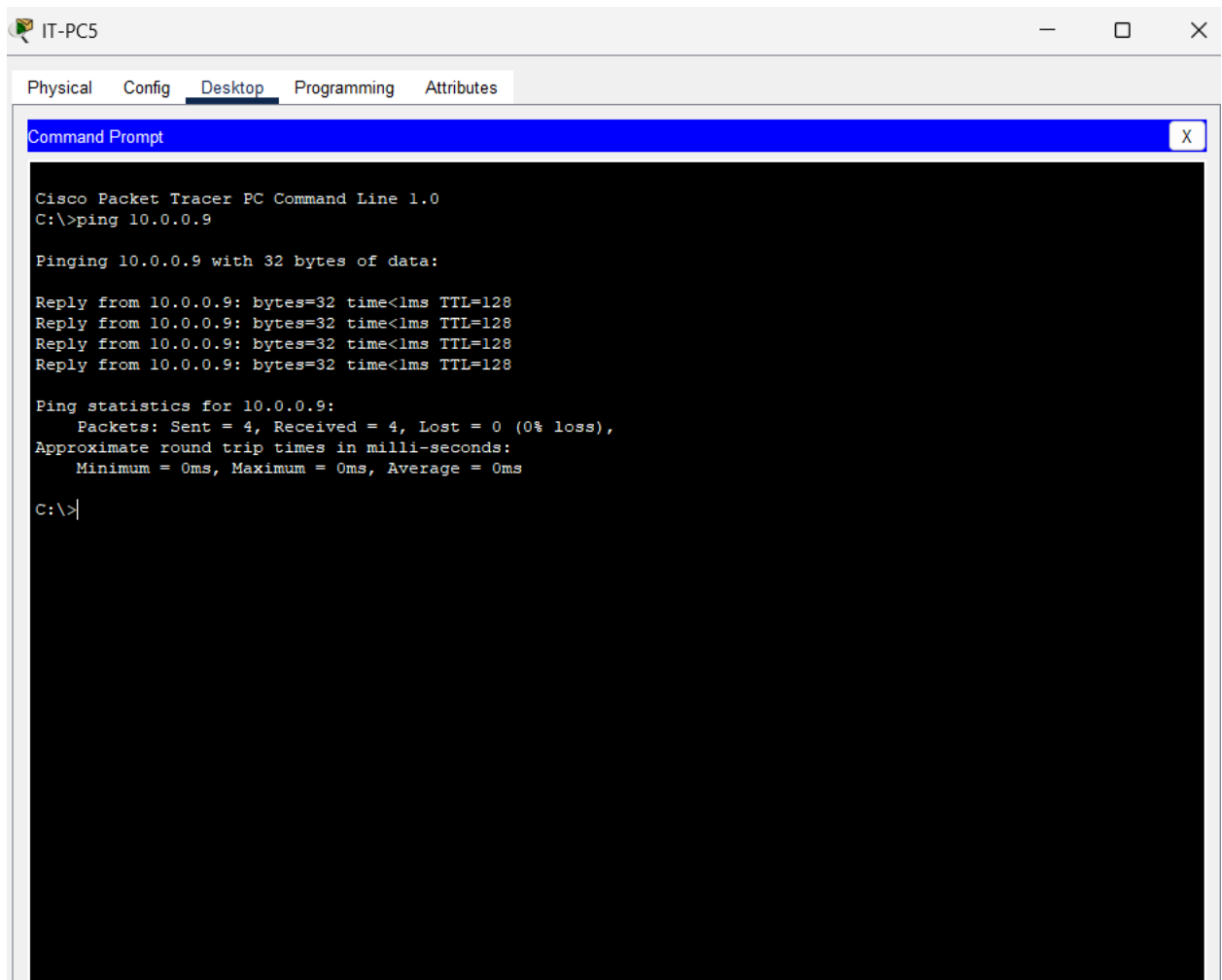
6. Testing and Verification

There is communication between the devices, both in the same department as well as on different departments.

A computer in admin department pings on in the legal department



A computer in the IT department pings another in the same department



The screenshot shows a window titled "IT-PC5" with a menu bar containing "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is selected, displaying a "Command Prompt" window. The command prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.9

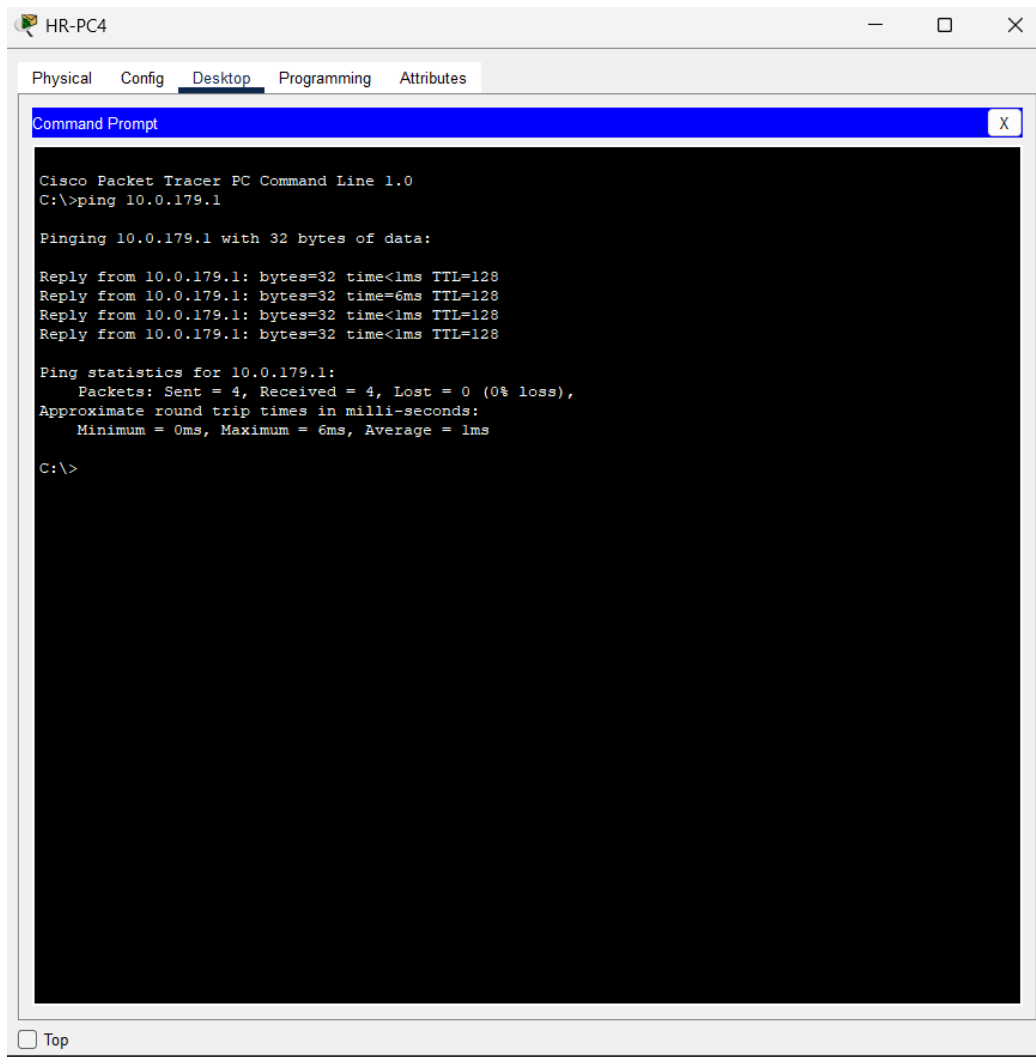
Pinging 10.0.0.9 with 32 bytes of data:

Reply from 10.0.0.9: bytes=32 time<1ms TTL=128
Reply from 10.0.0.9: bytes=32 time<1ms TTL=128
Reply from 10.0.0.9: bytes=32 time<1ms TTL=128
Reply from 10.0.0.9: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

The end devices are able to ping the server thus sending any request or data they may have



7. Expected behavior:

- Broadcast storm is to be expected since all the devices are on the same network bus cable, but the info will only be processed by the intended device before the message can be successfully replied to.

8. Advantages:

- It is easily expandable
- Cost-effective using a single cable line
- Management made simpler by the use of a single VLAN.

9. Limitations

- If the backbone of the network fails, the entire network is gone.
- Troubleshooting a bus topology is one of the most difficult topologies to decipher.
- All devices get info sent to them whether they are the recipient or not.

10. Conclusion:

It can finally be concluded that the project did in fact demonstrate all the expected behaviour of a typical bus topology while using Packet Tracer software. The daisy-chained access switches, common VLAN and all other devices used in the setup clearly showed that the network worked as a flat network and was efficient for the small department setup we did.